2024, Mar. 2024 Release LOC:Acute Adult
Infection: Pneumonia

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview

Instruction:
This subset is for patients with bacterial, fungal, or parasitic community- or healthcare-acquired pneumonia. For patients with known or highly suspected COVID-19, refer to the Infection: COVID-19 subset. For patients with actual or suspected viral respiratory infection, refer to the Infection: General subset. For patients who begin to exhibit signs and symptoms consistent with sepsis and need additional interventions which are not provided in the current subset, please refer to the Infection: Sepsis subset.

Introduction:
Pneumonia is defined as an acute infection of the lung parenchyma. Community-acquired pneumonia (CAP) refers to pneumonia that is acquired outside the hospital setting, whereas healthcare-acquired pneumonia (HAP) refers to pneumonia that occurs 48 hours or more after hospital admission. Clinical manifestations of CAP include cough, increased sputum production, dyspnea, fever, and/or pleuritic chest pain. The following physical exam findings have been reported to be independent predictors of CAP in adults: temperature of 37.8°C (100°F) or higher, crackles on auscultation, oxygen saturations less than 95%(0.95) and pulse greater than or equal to 100 beats/min. Older adults (65 years or older) and immunocompromised patients may exhibit more subtle, non-specific symptoms such as lethargy, change in mental status, dehydration, loss of appetite, and/or worsening of comorbidities, which may occur in the absence of fever. While the causal pathogen is often not identified, the most common bacterial pathogen isolated in CAP is streptococcus pneumoniae. Other common CAP pathogens include Haemophilus influenzae, Moraxella catarrhalis, and Staphylococcus aureus, and atypical organisms such as Legionella, Mycoplasma, and Chlamydophila pneumoniae (Aliberti et al., Lancet 2021, 398: 906-19). The microorganisms responsible for HAP differ based on geographic location; however, gram-negative bacilli such as Pseudomonas aeruginosa (P. aeruginosa) and Enterobacteriaceae are common causal pathogens in HAP (Luyt et al., Curr Opin Crit Care 2018, 24: 332-8).

Evaluation and Treatment:
The diagnosis of pneumonia is established based on the presenting symptoms, physical findings, and is confirmed by the presence of a new pulmonary infiltrate on imaging that is not due to another cause (National Institute for Health and Clinical Excellence (NICE): Pneumonia in adults: diagnosis and management. Clinical Guideline 191. 2019). Although a chest computerized tomography (CT) scan and lung ultrasonography can be used to diagnose pneumonia, a posteroanterior and lateral chest x-rays are the most used imaging techniques. Findings that may be seen on chest x-ray include focal non-segmental or lobar pneumonia, multifocal bronchopneumonia or lobular pneumonia, or focal or diffuse interstitial pneumonia. A chest CT may be needed to confirm pneumonia when there is high suspicion based on clinical findings if the chest x-ray is negative (e.g., in patients who are immunocompromised) (Aliberti et al., Lancet

2021, 398: 906-19).
Pretreatment diagnostic testing, such as gram stain and cultures of lower respiratory secretions and blood cultures, are not indicated except for the following patients (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67):

- Severe disease, especially if intubated
- Empirically treated for P. aeruginosa or methicillin-resistant Staph. aureus (MRSA)
- History of MRSA or P. aeruginosa
- Previously hospitalized and received parenteral antibiotics in the last 90 days (whether during hospitalization or not)

The use of procalcitonin to determine the need for initial antibacterial therapy is not recommended, as the threshold that differentiates between viral and bacterial pathogens in CAP is unknown (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67). Anti-infectives are the mainstay of treatment for patients with bacterial pneumonia. Empiric antibiotics selection should be made according to the most likely pathogen, local microbiology, patient risk factors for specific pathogens, and disease severity (Aliberti et al., Lancet 2021, 398: 906-19). Healthcare-acquired pneumonia is usually caused by gram-negative bacteria which are unique to each facility and initial treatment regimens are guided by local sensitivity patterns (Luyt et al., Curr Opin Crit Care 2018, 24: 332-8). If *Mycobacterium tuberculosis* is determined to be the cause, a multidrug anti-tuberculosis regimen is recommended (Moran and Waxman, In: Walls et al., ed. Rosen's Emergency Medicine: Concepts and Clinical Practice, 9th edn. 2018:871-80). Airborne isolation precautions are required until three consecutive negative AFB sputum smear or gastric aspirate results are obtained. Once clinically stable, discharge may be arranged (even when the smear remains positive) if the patient has begun a multidrug anti-tuberculosis regimen, does not pose a public health risk, and has directly observed therapy (DOT) arranged.
Intravenous antibiotics should be switched to an oral regimen as soon as there is clinical stability and the patient has been afebrile for 24 hours. In a patient with pneumonia, it may take up to 6 weeks for a CXR to show improvement. Follow up CXR should be limited to patients who are not improving or are demonstrating clinical deterioration (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67; Lim et al., Thorax 2009, 64 Suppl 3: iii1-55).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, AHRQ Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook.* The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The majority of the content in this subset is based on a limited number of key sources (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67; Lim et al., Thorax 2009; 64 Suppl 3: iii1-55).
Many pneumonia patients are at high risk for readmission, careful attention to discharge planning is essential.

(Excludes PO medications unless noted)

Select Day, One:

- **Initial review, One:** [1, 2, 3, 4, 5, 6]
  (From arrival in ED through decision to admit)
  (Symptom or finding within 24h)
  - **OBSERVATION, Both:** [7]
    - Pneumonia confirmed by imaging [8, 9, 10, 11]
    - Finding, ≥ **One:**
      - O_2 sat 89-91%(0.89-0.91) and < baseline [12]
      - Pneumonia Severity Index (PSI) Risk Class III (PSI score 71-90) [13, 14]
      - CURB-65 criteria, **Two:** [14, 15]
        - New onset confusion [16]
        - Blood urea nitrogen (BUN) ≥ 20 mg/dL(7 mmol/L)
        - Respiratory rate ≥ 30/min
        - Systolic blood pressure < 90 mmHg or diastolic blood pressure ≤ 60 mmHg
        - Age ≥ 65 years
      - Failed outpatient anti-infective (includes PO), ≥ **One:** [17, 18]
        - Continued deterioration despite ≥ 24h anti-infective treatment [19]
        - Noncompliant with outpatient anti-infective
        - Unable to tolerate or absorb oral anti-infective
        - Unresponsive to outpatient anti-infective, ≥ **One:**
          - ≥ 3d
          - ≥ 5 doses
  - **ACUTE, Both:** [20]
    - Pneumonia confirmed by imaging [8, 9, 10, 11]
    - Finding, ≥ **One:**
      - O_2 sat < 89%(0.89) and < baseline [12]
      - Arterial Po_2 < 56 mmHg(7.4 kPa) and < baseline [12]
      - Arterial or venous Pco_2 45-54 mmHg(6.0-7.2 kPa) and pH 7.31-7.35 [21]
      - Complicated pneumonia [22, 23, 24]
      - Frailty, **All:** [25, 26]
        - Age ≥ 65 yrs
        - Finding, ≥ **One:**
          - Frailty, active diagnosis [27]
          - Frailty characteristics, ≥ **Two:** [28]
            - Chronic condition(s), ≥ **One:**
              - Dementia or cognitive impairment [29]
              - Multimorbidity, ≥ 2 chronic illnesses [30, 31]
            - Functional impairment [32, 33]
            - Malnutrition [34]
            - Polypharmacy, ≥ 5 chronic medications [35, 36]
        - Symptom, ≥ **One:**
          - O_2 sat 89-91%(0.89-0.91) and < baseline [12]
          - Pneumonia Severity Index (PSI) Risk Class III (PSI score 71-90) [13, 14]
          - CURB-65 criteria, ≥ **One:** [14, 15]
            - New onset confusion [16]
            - Blood urea nitrogen (BUN) ≥ 20 mg/dL(7 mmol/L)
            - Respiratory rate ≥ 30/min
            - Systolic blood pressure < 90 mmHg or diastolic blood pressure ≤ 60 mmHg
      - High risk for adverse event, **Both:**
        - O_2 sat 89-91%(0.89-0.91) and < baseline [12]
        - Clinical risk factor, ≥ **One:** [37]
          - COPD, Grade 3 or 4 (current) [38]

- Heart failure, NYHA Class III or IV (current) (39, 40)
- Severe and persistent mental illness or substance use disorder and, **Both:** (41, 42)
  - Patient unreliable and unable to adhere to outpatient instructions or treatment (43)
  - Caregiver unavailable or unable to manage care
- Immunocompromised (44, 45)
- Pneumonia Severity Index (PSI) Risk Class IV or V (PSI score ≥ 91) (13, 14)
- CURB-65 criteria, ≥ **Three:** (14, 15)
  - New onset confusion (16)
  - Blood urea nitrogen (BUN) ≥ 20 mg/dL(7 mmol/L)
  - Respiratory rate ≥ 30/min
  - Systolic blood pressure < 90 mmHg or diastolic blood pressure ≤ 60 mmHg
  - Age ≥ 65 years

INTERMEDIATE, One: (46)

- Pneumonia confirmed by imaging and, ≥ **One:** (8, 9, 10, 11)
  - IDSA/ATS minor criteria for severe community-acquired pneumonia, ≥ **Three:** (47)
    - Respiratory rate ≥ 30/min
    - PaO_2/FiO_2 ratio ≤ 250
    - Multilobar infiltrates
    - BUN ≥ 20.0 mg/dL(7 mmol/L)
    - WBC < 4,000/cu.mm(4 x10^9/L) (48)
    - Platelet count < 100,000/cu.mm(100x10^9/L)
    - Temperature < 96.8°F(36°C)
    - Hypotension requiring fluid resuscitation (49)
  - HFNC and respiratory monitoring every 3-4h (50, 51, 52)
  - NIPPV (53)
  - Oxygen ≥ 40%(0.40), ≤ 2d (54)

CRITICAL, One: (55)

- Pneumonia confirmed by imaging and, ≥ **One:** (8, 9, 10, 11)
  - ECMO or ECLS (56)
  - Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, **Both:** (57)
    - Hypotension, ≥ **One:**
      - MAP < 65 mmHg
      - Systolic blood pressure < 90 mmHg and < baseline (12)
    - Finding, ≥ **One:**
      - Lactic acid > 2 mmol/L(18.0 mg/dL) (58)
      - Urine output < 0.5 mL/kg/h
  - Hemodynamic monitoring, invasive (59)
  - HFNC and respiratory monitoring every 1-2h (50, 51, 52, 54)
  - Mechanical ventilation
  - NIPPV and, ≥ **One:** (53)
    - Arterial Po_2 ≤ 39 mmHg(5.2 kPa)
    - Arterial or venous Pco_2 ≥ 60 mmHg(8.0 kPa) and, **One:**
      - COPD and arterial or venous pH ≤ 7.24
      - No hx of COPD
  - Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

Episode Day 1, One: (1, 2, 4, 5, 6)

(Symptom or finding within 24h)

OBSERVATION, One: (7)

- Pneumonia confirmed by imaging and, **Both:** (8, 11)
  - Finding, ≥ **One:**
    - O_2 sat 89-91%(0.89-0.91) and < baseline (12)
    - Pneumonia Severity Index (PSI) Risk Class III (PSI score 71-90) (13, 14)

- CURB-65 criteria, **Two:** (14, 15)
  - New onset confusion (16)
  - Blood urea nitrogen (BUN) ≥ 20 mg/dL(7 mmol/L)
  - Respiratory rate ≥ 30/min
  - Systolic blood pressure < 90 mmHg or diastolic blood pressure ≤ 60 mmHg
  - Age ≥ 65 years
- Failed outpatient anti-infective (includes PO), ≥ **One:** (17, 18)
  - Continued deterioration despite ≥ 24h anti-infective treatment (19)
  - Noncompliant with outpatient anti-infective
  - Unable to tolerate or absorb oral anti-infective
  - Unresponsive to outpatient anti-infective, ≥ **One:**
    - ≥ 3d
    - ≥ 5 doses

- Intervention, ≥ **One:** (10)
  - Anti-infective (9)
  - PO anti-infective and, ≥ **One:** (9)
    - IV fluid (60)
    - Requiring supplemental oxygen (54)

- **ACUTE, One:** (20)
  - Pneumonia confirmed by imaging and, **Both:** (8, 11)
    - Finding, ≥ **One:**
      - O_2 sat < 89%(0.89) and < baseline (12)
      - Arterial Po_2 < 56 mmHg(7.4 kPa) and < baseline (12)
      - Arterial or venous Pco_2 45-54 mmHg(6.0-7.2 kPa) and pH 7.31-7.35 (21)
      - Complicated pneumonia (22, 23, 24)
      - Frailty, **All:** (25, 26)
        - Age ≥ 65 yrs
        - Finding, ≥ **One:**
          - Frailty, active diagnosis (27)
          - Frailty characteristics, ≥ **Two:** (28)
            - Chronic condition(s), ≥ **One:**
              - Dementia or cognitive impairment (29)
              - Multimorbidity, ≥ 2 chronic illnesses (30, 31)
            - Functional impairment (32, 33)
            - Malnutrition (34)
            - Polypharmacy, ≥ 5 chronic medications (35, 36)
        - Symptom, ≥ **One:**
          - O_2 sat 89-91%(0.89-0.91) and < baseline (12)
          - Pneumonia Severity Index (PSI) Risk Class III (PSI score 71-90) (13, 14)
          - CURB-65 criteria, ≥ **One:** (14, 15)
            - New onset confusion (16)
            - Blood urea nitrogen (BUN) ≥ 20 mg/dL(7 mmol/L)
            - Respiratory rate ≥ 30/min
            - Systolic blood pressure < 90 mmHg or diastolic blood pressure ≤ 60 mmHg
      - High risk for adverse event, **Both:**
        - O_2 sat 89-91%(0.89-0.91) and < baseline (12)
        - Clinical risk factor, ≥ **One:** (37)
          - COPD, Grade 3 or 4 (current) (38)
          - Heart failure, NYHA Class III or IV (current) (39, 40)
          - Severe and persistent mental illness or substance use disorder and, **Both:** (41, 42)
            - Patient unreliable and unable to adhere to outpatient instructions or treatment (43)
            - Caregiver unavailable or unable to manage care
      - Immunocompromised (44, 45)

- Pneumonia Severity Index (PSI) Risk Class IV or V (PSI score ≥ 91) [13, 14]
- ⊖ CURB-65 criteria, ≥ **Three:** [14, 15]
  - New onset confusion [16]
  - Blood urea nitrogen (BUN) ≥ 20 mg/dL(7 mmol/L)
  - Respiratory rate ≥ 30/min
  - Systolic blood pressure < 90 mmHg or diastolic blood pressure ≤ 60 mmHg
  - Age ≥ 65 years
- Comorbid pneumonia in a hospitalized patient confirmed by imaging [8, 11, 61]

⊖ Intervention, ≥ **One:** [10]
- Anti-infective [9]
- ⊖ PO anti-infective and, ≥ **One:** [9]
  - IV fluid [60]
  - Requiring supplemental oxygen [54]

⊖ **INTERMEDIATE, One:** [46]
- ⊖ Pneumonia confirmed by imaging and, ≥ **One:** [8, 9, 10, 11]
  - ⊖ IDSA/ATS minor criteria for severe community-acquired pneumonia, ≥ **Three:** [47]
    - Respiratory rate ≥ 30/min
    - PaO_2/FiO_2 ratio ≤ 250
    - Multilobar infiltrates
    - BUN ≥ 20.0 mg/dL(7 mmol/L)
    - WBC < 4,000/cu.mm(4 x10^9/L) [48]
    - Platelet count < 100,000/cu.mm(100x10^9/L)
    - Temperature < 96.8°F(36°C)
    - Hypotension requiring fluid resuscitation [49]
  - HFNC and respiratory monitoring every 3-4h [50, 51, 52]
  - NIPPV [53]
  - Oxygen ≥ 40%(0.40), ≤ 2d [54]

⊖ **CRITICAL, One:** [55]
- ⊖ Pneumonia confirmed by imaging and, ≥ **One:** [8, 9, 10, 11]
  - ECMO or ECLS [56]
  - ⊖ Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, **Both:** [57]
    - ⊖ Hypotension, ≥ **One:**
      - MAP < 65 mmHg
      - Systolic blood pressure < 90 mmHg and < baseline [12]
    - ⊖ Finding, ≥ **One:**
      - Lactic acid > 2 mmol/L(18.0 mg/dL) [58]
      - Urine output < 0.5 mL/kg/h
  - Hemodynamic monitoring, invasive [59]
  - HFNC and respiratory monitoring every 1-2h [50, 51, 52, 54]
  - Mechanical ventilation
  - ⊖ NIPPV and, ≥ **One:** [53]
    - Arterial Po_2 ≤ 39 mmHg(5.2 kPa)
    - ⊖ Arterial or venous Pco_2 ≥ 60 mmHg(8.0 kPa) and, **One:**
      - COPD and arterial or venous pH ≤ 7.24
      - No hx of COPD
  - Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

⊖ **Episode Day 2, One:** [4, 5, 62]
- ⊖ **OBSERVATION, One:**
  - ⊖ **Responder**, discharge expected today if clinically stable last 12h, **All:** [63, 64]
    - ⊖ Anti-infective, **One:**
      - Arranged
      - Completed

- Not indicated
- O_2 sat within acceptable limits [65]
- Laboratory values normal or within acceptable limits [65]
- Tolerating PO [66]

- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
  - Pneumonia and, ≥ **One:**
    - Anti-infective [9]
    - PO anti-infective and, **Both:** [9]
      - O_2 sat < 92%(0.92) and < baseline [12]
      - Requiring supplemental oxygen [54]
  - Inadequate oral intake and IV fluid [60, 67]

- **ACUTE, One:**
  - **Early responder**, discharge expected today if clinically stable last 12h, **All:** [63, 64]
    - Anti-infective, **One:**
      - Arranged
      - Completed
      - Not indicated
    - O_2 sat within acceptable limits [65]
    - Laboratory values normal or within acceptable limits [65]
    - Tolerating PO or nutritional route established [66, 68]
    - No complication or active comorbidity relevant to this episode of care [69]
  - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
    - Pneumonia and, ≥ **One:**
      - Anti-infective [9]
      - PO anti-infective and, ≥ **One:** [9, 70]
        - O_2 sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen [12, 54]
        - Respiratory intervention ≥ 4x/24h [71]
      - Chest tube placement within last 24h [72]
    - Aspiration pneumonitis, actual or suspected and, ≥ **One:**
      - Assessment of swallowing function ≤ 24h [73]
      - Feeding tube insertion within last 24h
    - COPD and, **All:**
      - Finding, ≥ **One:**
        - O_2 sat ≤ 91%(0.91) and < baseline [12]
        - Arterial Po_2 ≤ 60 mmHg(8.0 kPa) [21]
        - Arterial or venous Pco_2 > baseline [12, 21]
      - Bronchodilator ≥ 4x/24h [74]
      - Corticosteroid (includes PO or inhaled)
      - Oxygenation, **One:**
        - O_2 sat ≥ 92%(0.92) or baseline [12]
        - O_2 sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen [12, 54]
    - Functional impairment (new) and rehabilitation initiation ≤ 24h [75, 76, 77]
    - Heart failure and, **Both:** [78]
      - Dyspnea > baseline and, ≥ **One:** [12]
        - O_2 sat ≤ 91%(0.91) and < baseline [12]
        - Worsening edema
        - Heart failure confirmed by imaging [79]
      - Intervention, ≥ **One:** [80]
        - Diuretic, **One:** [81]
          - ≥ 2x/24h (includes PO)
          - At least 1x/24h and requires daily medication adjustment (includes PO) and, ≥ **One:** [82]
            - Blood urea nitrogen (BUN) increased from baseline [12]
            - Creatinine increased from baseline [12]

- Dialysis
- Inadequate oral intake and IV fluid [60, 67]
- Post Critical care ≤ 24h
- **INTERMEDIATE, ≥ One:**
  - HFNC and respiratory monitoring every 3-4h [50, 51, 52]
  - NIPPV [53]
  - Oxygen ≥ 40%(0.40), ≤ 2d [54]
  - Volume expander, ≥ **One:** [83, 84]
    - Crystalloid ≥ 30 ml/kg/24h or colloid equivalent
    - Crystalloid ≥ 2 L/24h or colloid equivalent
- **CRITICAL, ≥ One:**
  - ECMO or ECLS [56]
  - Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, **Both:** [57]
    - Hypotension, ≥ **One:**
      - MAP < 65 mmHg
      - Systolic blood pressure < 90 mmHg and < baseline [12]
    - Finding, ≥ **One:**
      - Lactic acid > 2 mmol/L(18.0 mg/dL) [58]
      - Urine output < 0.5 mL/kg/h
  - Hemodynamic monitoring, invasive [59]
  - HFNC and respiratory monitoring every 1-2h [50, 51, 52, 54]
  - Mechanical ventilation
  - NIPPV and, ≥ **One:** [53]
    - FiO_2 > 50%(0.50)
    - Respiratory intervention every 1-2h [71]
  - Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

Episode Day 3, One: [5, 85, 86]

- **OBSERVATION, One:**
  - **Responder**, discharge expected today if clinically stable last 12h, **All:** [64]
    - Anti-infective, **One:**
      - Arranged
      - Completed
      - Not indicated
    - O_2 sat within acceptable limits [12]
    - Laboratory values normal or within acceptable limits [65]
    - Tolerating PO [66]
  - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
    - For pneumonia use Episode Day 3 at a higher level of care
    - For a condition other than pneumonia use appropriate subset (Episode Day 1) based on clinical findings
    - Refer for secondary review
- **ACUTE, One:**
  - **Responder**, discharge expected today if clinically stable last 12h, **All:** [63, 64]
    - Anti-infective, **One:**
      - Arranged
      - Completed
      - Not indicated
    - Laboratory values normal or within acceptable limits [65]
    - O_2 sat within acceptable limits [65]
    - Tolerating PO or nutritional route established [66, 68]
    - Complication or comorbidity, **One:**
      - No complication or active comorbidity relevant to this episode of care [69]

- Complication or comorbidity and clinically stable for discharge, ≥ **One:**
  - Aspiration pneumonitis and, **One:**
    - Able to swallow without aspirating
    - Feeding tube insertion
  - Chest tube and, **One:** [72]
    - Heimlich valve
    - Removed and no reaccumulation of air or fluid confirmed by chest x-ray (CXR)
  - COPD and, ≥ **One:**
    - O_2 sat within acceptable limits [65]
    - Arterial Po_2 > 60 mmHg(8.0 kPa) or within acceptable limits [21, 65]
    - Arterial or venous Pco_2 within acceptable limits [21, 65]
  - Functional evaluation complete [75, 87, 88]
  - Heart failure resolving and, **Both:**
    - Dyspnea resolving or at baseline [12]
    - O_2 sat within acceptable limits [65]
- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
  - Pneumonia and, ≥ **One:**
    - Anti-infective [9]
    - PO anti-infective and, ≥ **One:** [9, 70]
      - O_2 sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen [12, 54]
      - Respiratory intervention ≥ 4x/24h [71]
  - Aspiration pneumonitis, actual or suspected and, ≥ **One:**
    - Assessment of swallowing function ≤ 24h [73]
    - Feeding tube insertion within last 24h
  - Chest tube and, ≥ **One:** [72]
    - Suction, continuous
    - Drainage ≥ 250 mL/24h
    - Repositioning within last 24h
    - Water seal within last 24h
  - COPD and, **All:**
    - Finding, ≥ **One:**
      - O_2 sat ≤ 91%(0.91) and < baseline [12]
      - Arterial Po_2 ≤ 60 mmHg(8.0 kPa) [21]
      - Arterial or venous Pco_2 > baseline [12, 21]
    - Bronchodilator ≥ 4x/24h [74]
    - Corticosteroid (includes PO or inhaled)
    - Oxygenation, **One:**
      - O_2 sat ≥ 92%(0.92) or baseline [12]
      - O_2 sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen [12, 54]
  - Functional impairment (new) and rehabilitation initiation ≤ 24h [75, 76, 77]
  - Heart failure and, **Both:** [78]
    - Dyspnea > baseline and, ≥ **One:** [12]
      - O_2 sat ≤ 91%(0.91) and < baseline [12]
      - Worsening edema
      - Heart failure confirmed by imaging [79]
    - Intervention, ≥ **One:** [80]
      - Diuretic, **One:** [81]
        - ≥ 2x/24h (includes PO)
        - At least 1x/24h and requires daily medication adjustment (includes PO) and, ≥ **One:** [82]
          - Blood urea nitrogen (BUN) increased from baseline [12]
          - Creatinine increased from baseline [12]
      - Dialysis
  - Inadequate oral intake and IV fluid [60, 67]

- − Post Critical care ≤ 24h
- ⊖ **INTERMEDIATE,** ≥ **One:**
  - − HFNC and respiratory monitoring every 3-4h [50, 51, 52]
  - − NIPPV [53]
  - − Oxygen ≥ 40%(0.40), ≤ 2d [54]
  - ⊖ Volume expander, ≥ **One:** [83, 84]
    - − Crystalloid ≥ 30 ml/kg/24h or colloid equivalent
    - − Crystalloid ≥ 2 L/24h or colloid equivalent
- ⊖ **CRITICAL,** ≥ **One:**
  - − ECMO or ECLS [56]
  - ⊖ Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, **Both:** [57]
    - ⊖ Hypotension, ≥ **One:**
      - − MAP < 65 mmHg
      - − Systolic blood pressure < 90 mmHg and < baseline [12]
    - ⊖ Finding, ≥ **One:**
      - − Lactic acid > 2 mmol/L(18.0 mg/dL) [58]
      - − Urine output < 0.5 mL/kg/h
  - − Hemodynamic monitoring, invasive [59]
  - − HFNC and respiratory monitoring every 1-2h [50, 51, 52, 54]
  - − Mechanical ventilation
  - ⊖ NIPPV and, ≥ **One:** [53]
    - − FiO_2 > 50%(0.50)
    - − Respiratory intervention every 1-2h [71]
  - − Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

⊖ **Episode Day 4, One:** [5, 86]

- ⊖ **ACUTE, One:**
  - ⊖ **Responder**, discharge expected today if clinically stable last 12h, **All:** [63, 64]
    - ⊖ Anti-infective, **One:**
      - − Arranged
      - − Completed
      - − Not indicated
    - − Laboratory values normal or within acceptable limits [65]
    - − O_2 sat within acceptable limits [65]
    - − Tolerating PO or nutritional route established [66, 68]
    - ⊖ Complication or comorbidity, **One:**
      - − No complication or active comorbidity relevant to this episode of care [69]
      - ⊖ Complication or comorbidity and clinically stable for discharge, ≥ **One:**
        - ⊖ Aspiration pneumonitis and, **One:**
          - − Able to swallow without aspirating
          - − Feeding tube insertion
        - ⊖ Chest tube and, **One:** [72]
          - − Heimlich valve
          - − Removed and no reaccumulation of air or fluid confirmed by chest x-ray (CXR)
        - ⊖ COPD and, ≥ **One:**
          - − O_2 sat within acceptable limits [65]
          - − Arterial Po_2 > 60 mmHg(8.0 kPa) or within acceptable limits [21, 65]
          - − Arterial or venous Pco_2 within acceptable limits [21, 65]
        - − Functional evaluation complete [75, 87, 88]
        - ⊖ Heart failure resolving and, **Both:**
          - − Dyspnea resolving or at baseline [12]
          - − O_2 sat within acceptable limits [65]
  - ⊖ **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**

- ⊖ Pneumonia and, ≥ **One:**
  - − Anti-infective [9]
  - ⊖ PO anti-infective and, ≥ **One:** [9, 70]
    - − O_2 sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen [12, 54]
    - − Respiratory intervention ≥ 4x/24h [71]
- ⊖ Aspiration pneumonitis, actual or suspected and, ≥ **One:**
  - − Assessment of swallowing function ≤ 24h [73]
  - − Feeding tube insertion within last 24h
- ⊖ Chest tube and, ≥ **One:** [72]
  - − Suction, continuous
  - − Drainage ≥ 250 mL/24h
  - − Repositioning within last 24h
  - − Water seal within last 24h
- ⊖ COPD and, **All:**
  - ⊖ Finding, ≥ **One:**
    - − O_2 sat ≤ 91%(0.91) and < baseline [12]
    - − Arterial Po_2 ≤ 60 mmHg(8.0 kPa) [21]
    - − Arterial or venous Pco_2 > baseline [12, 21]
  - − Bronchodilator ≥ 4x/24h [74]
  - − Corticosteroid (includes PO or inhaled)
  - ⊖ Oxygenation, **One:**
    - − O_2 sat ≥ 92%(0.92) or baseline [12]
    - − O_2 sat ≤ 91%(0.91) and < baseline requiring supplemental oxygen [12, 54]
- − Functional impairment (new) and rehabilitation initiation ≤ 24h [75, 76, 77]
- ⊖ Heart failure and, **Both:** [78]
  - ⊖ Dyspnea > baseline and, ≥ **One:** [12]
    - − O_2 sat ≤ 91%(0.91) and < baseline [12]
    - − Worsening edema
    - − Heart failure confirmed by imaging [79]
  - ⊖ Intervention, ≥ **One:** [80]
    - ⊖ Diuretic, **One:** [81]
      - − ≥ 2x/24h (includes PO)
      - ⊖ At least 1x/24h and requires daily medication adjustment (includes PO) and, ≥ **One:** [82]
        - − Blood urea nitrogen (BUN) increased from baseline [12]
        - − Creatinine increased from baseline [12]
    - − Dialysis
- − Inadequate oral intake and IV fluid [60, 67]
- − Post Critical care ≤ 24h

⊖ **INTERMEDIATE,** ≥ **One:**
- − HFNC and respiratory monitoring every 3-4h [50, 51, 52]
- − NIPPV [53]
- − Oxygen ≥ 40%(0.40), ≤ 2d [54]
- ⊖ Volume expander, ≥ **One:** [83, 84]
  - − Crystalloid ≥ 30 ml/kg/24h or colloid equivalent
  - − Crystalloid ≥ 2 L/24h or colloid equivalent

⊖ **CRITICAL,** ≥ **One:**
- − ECMO or ECLS [56]
- ⊖ Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, **Both:** [57]
  - ⊖ Hypotension, ≥ **One:**
    - − MAP < 65 mmHg
    - − Systolic blood pressure < 90 mmHg and < baseline [12]
  - ⊖ Finding, ≥ **One:**

- Lactic acid > 2 mmol/L(18.0 mg/dL) [58]
- Urine output < 0.5 mL/kg/h
- Hemodynamic monitoring, invasive [59]
- HFNC and respiratory monitoring every 1-2h [50, 51, 52, 54]
- Mechanical ventilation
- ⊖ NIPPV and, ≥ **One:** [53]
  - FiO_2 > 50%(0.50)
  - Respiratory intervention every 1-2h [71]
- Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

- ⊖ **Episode Day 5, One:**
  - ⊖ **ACUTE, One:**
    - ⊖ **Responder**, discharge expected today if clinically stable last 12h, **All:** [63, 64]
      - ⊖ Anti-infective, **One:**
        - Arranged
        - Completed
        - Not indicated
      - Laboratory values normal or within acceptable limits [65]
      - O_2 sat within acceptable limits [65]
      - Tolerating PO or nutritional route established [66, 68]
      - ⊖ Complication or comorbidity, **One:**
        - No complication or active comorbidity relevant to this episode of care [69]
        - ⊖ Complication or comorbidity and clinically stable for discharge, ≥ **One:**
          - ⊖ Aspiration pneumonitis and, **One:**
            - Able to swallow without aspirating
            - Feeding tube insertion
          - ⊖ Chest tube and, **One:** [72]
            - Heimlich valve
            - Removed and no reaccumulation of air or fluid confirmed by chest x-ray (CXR)
          - ⊖ COPD and, ≥ **One:**
            - O_2 sat within acceptable limits [65]
            - Arterial Po_2 > 60 mmHg(8.0 kPa) or within acceptable limits [21, 65]
            - Arterial or venous Pco_2 within acceptable limits [21, 65]
          - Functional evaluation complete [75, 87, 88]
          - ⊖ Heart failure resolving and, **Both:**
            - Dyspnea resolving or at baseline [12]
            - O_2 sat within acceptable limits [65]
    - ⊖ **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
      - **Use Extended Stay subset**
      - Refer for secondary review
  - ⊖ **INTERMEDIATE, One:**
    - ⊖ **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**
      - **Use Extended Stay subset**
      - Refer for secondary review
  - ⊖ **CRITICAL, One:**
    - ⊖ **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
      - **Use Extended Stay subset**
      - Refer for secondary review

Discharge, Level of Care, One: [89]

- ⊖ **HOME, Both:**

- Level of care appropriateness, **Both:**
  - Home environment safe and accessible [90]
  - Patient or caregiver demonstrates ability to manage care
- Pneumonia discharge planning, **All:**
  - Provide comprehensive written discharge and teaching instructions [91]
  - Education on the importance of, **Both:**
    - Smoking cessation, if indicated [92]
    - Influenza or pneumococcal vaccine [93]
  - Medication reconciliation [94]
  - Follow-up care planned [95]
  - Identify and address transportation needs

- **HOME CARE, Both:**
  - Level of care appropriateness, **All:**
    - Home environment safe and accessible [90]
    - Patient or caregiver willing and able to learn care
    - Outpatient management contraindicated or unavailable [96, 97]
    - Skilled services, ≥ **One:** [98]
      - Skilled nursing [99]
      - Skilled therapy, PT, OT, or ST
      - Behavioral health [100]
  - Pneumonia discharge planning, **All:**
    - Provide comprehensive written discharge and teaching instructions [91]
    - Education on the importance of, **Both:**
      - Smoking cessation, if indicated [92]
      - Influenza or pneumococcal vaccine [93]
    - Medication reconciliation [94]
    - Follow-up care planned [95]
    - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
  - Level of care appropriateness, **One:**
    - Support systems available [101]
  - Skilled treatment, ≥ **One:**
    - Clinical assessment [102]
    - Individual, group, or family counseling
    - Psychiatric, substance, or medication evaluation [103]
- **SKILLED MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
    - Treatment precluded in a lower level
    - Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 1-2x/24h [99]
      - Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [104, 105]
        - Functional impairments requiring at least supervision [106]
        - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [94]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk

- Treatment precluded in a lower level
- Services, ≥ **One:**
  - Medical and skilled nursing interventions or assessment 4h/24h [99]
  - Therapy, **All:**
    - Goal directed therapy with rehabilitation potential [104, 105]
    - ≥ 2 functional impairments requiring at least minimum assistance [106]
    - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
- Complete prior to facility transfer, **All:**
  - Medication reconciliation [94]
  - Obtain and complete forms for facility
  - Obtain discharge summary and transmit to facility and medical practitioner
  - Arrange transportation

- **SUBACUTE REHAB, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Treatment precluded in a lower level
    - Services, **All:**
      - Goal directed therapy with rehabilitation potential [104, 105]
      - ≥ 2 functional impairments requiring at least minimum assistance [106]
      - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [94]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **ACUTE REHAB, Both:**
  - Level of care appropriateness, **Both:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Services, **All:**
      - Goal directed therapy with rehabilitation potential [104, 105]
      - ≥ 2 functional impairments requiring at least minimum assistance [106]
      - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines [107]
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [94]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
  - Level of care appropriateness, **Both:**
    - Treatment precluded in a lower level
    - In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One:**
      - Primary condition or illness and treatment of active comorbid condition(s) [108]
      - Ventilator dependent ≥ 6h/d and weaning planned [109, 110]
      - Acute and comorbid conditions requiring prolonged hospitalization [111]
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [94]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
177 RESPIRATORY INFECTIONS AND INFLAMMATIONS WITH MCC	n/a	5	CMS GMLOS
178 RESPIRATORY INFECTIONS AND INFLAMMATIONS WITH CC	n/a	3.3	CMS GMLOS
179 RESPIRATORY INFECTIONS AND INFLAMMATIONS WITHOUT CC/MCC	n/a	2.5	CMS GMLOS
193 SIMPLE PNEUMONIA AND PLEURISY WITH MCC	n/a	4.1	CMS GMLOS
194 SIMPLE PNEUMONIA AND PLEURISY WITH CC	n/a	2.9	CMS GMLOS
195 SIMPLE PNEUMONIA AND PLEURISY WITHOUT CC/MCC	n/a	2.4	CMS GMLOS
196 INTERSTITIAL LUNG DISEASE WITH MCC	n/a	5.2	CMS GMLOS
197 INTERSTITIAL LUNG DISEASE WITH CC	n/a	3	CMS GMLOS
198 INTERSTITIAL LUNG DISEASE WITHOUT CC/MCC	n/a	2.3	CMS GMLOS
202 BRONCHITIS AND ASTHMA WITH CC/MCC	n/a	2.9	CMS GMLOS
203 BRONCHITIS AND ASTHMA WITHOUT CC/MCC	n/a	2.2	CMS GMLOS
205 OTHER RESPIRATORY SYSTEM DIAGNOSES WITH MCC	n/a	4.5	CMS GMLOS
206 OTHER RESPIRATORY SYSTEM DIAGNOSES WITHOUT MCC	n/a	2.5	CMS GMLOS
867 OTHER INFECTIOUS AND PARASITIC DISEASES DIAGNOSES WITH MCC	n/a	5.5	CMS GMLOS
868 OTHER INFECTIOUS AND PARASITIC DISEASES DIAGNOSES WITH CC	n/a	3.7	CMS GMLOS
869 OTHER INFECTIOUS AND PARASITIC DISEASES DIAGNOSES WITHOUT CC/MCC	n/a	2.5	CMS GMLOS
PNEUMONIA	16%	4.3	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:

1:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.

- **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Transition Plan: InterQual's® Transition Plan identifies patients at high risk for readmission who may benefit from a comprehensive discharge plan. Factors that have been associated with high risk for 30-day readmission after hospitalization for pneumonia include male sex assigned at birth and the following chronic comorbid conditions: cancer, heart failure, chronic respiratory disease, chronic kidney disease, and diabetes mellitus (Fang et al., World J Clin Cases 2022, 10: 3787-800).

3:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory,

imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.
Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.
Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and intensity of service requirements.
For further clarification, see the Acute Level of Care Review Process.

4:

Expected progress after 24-48h
- Respiratory rate improving
- O_2 requirement decreasing
- O_2 saturation improving
- Activity increasing

Consider:
- Changing antibiotic route from parenteral to PO
- Physical therapy consult for ambulation and prevention of deconditioning
- Discharge planning

5:

Care Facilitation
Prolonged oxygen therapy expected
- Home oxygen therapy should be considered in a clinically stable patient who continues to require supplemental oxygen when there has been no change in oxygen requirements within last 2 days **(Home)**

Potential risk of readmission
- Knowledge deficit or non-adherence with condition management identified **(Home Care, Readmission reduction team)**
- Co-morbid chronic obstructive pulmonary disease or heart failure **(Home Care, Readmission reduction team)**

Functional or ADL impairment - consider an alternate level of care for therapy needs
- Physical therapy provided in the home setting should be considered if the home is safe and accessible **(Home Care)**
- If home environment is unsafe, consider a skilled nursing facility **(SNF)**

Prolonged ventilation and unable to wean
- Prolonged ventilator weaning expected - consider transfer to a LTAC for failed mechanical ventilation weaning longer than 1 week (**SAC, LTAC)**

Social Determinants of Health (SDOH)
Social concerns may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. The following social determinants of health should be evaluated on admission to determine which factors are relevant to an individual patient and condition. This list may not be all inclusive and all relevant SDOH factors should be considered. Any SDOH factors determined to be relevant to a given patient should be planned for throughout the hospital stay.
Alcohol or substance use
- Care coordination with outpatient medical or behavioral health providers
- Information should be given about community resources and programs
- Referral to addiction social support systems

Domestic violence and sexually abused individuals
- Community resources and legal information should be given
- Create a long-term safety plan or make appropriate referral

End of life planning, arrange palliative care or hospice

- Consider palliative care for patients with end-stage disease
- Consider hospice care for terminally ill patients with a life expectancy of less than 6 months

Financial resources strained or limited

- For patients with income insecurity, identify income-support resources (Pottie et al., CMAJ 2020, 192: E240-E54)
- Consider direct income assistance benefits and indirect income assistance programs (i.e., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)
- Consider referral to Area Agencies on Aging (AAA) for information on nutrition and meals programs, homemaking services, transportation resources
- For food insecurity, provide information about local food banks and Supplemental Nutrition Assistance Program (SNAP) incentive programs (Berman et al., Pediatr Rev 2018, 39: 235-46)

Functional or ADL impairment

- Consider an alternate level of care
- If patient is unable to make decision for self, refuses treatment, or demands to be discharged without a rational reason, a capacity evaluation may be needed to ensure patient's ability to make decisions for self
- Information should be given to patient and family about Advance Directive or Power of Attorney

Mental health co-morbidities involved

- Appointment with primary care provider or behavioral health specialist upon discharge
- Care coordination with outpatient behavioral health providers
- Community resources should be given

Risk for readmission

- Consider home care services
- Consider transfer to an alternative level of care or readmission reduction program
- Provide patient and family with information on Disease Management programs
- Consider telehealth and mobile health interventions

Risk for treatment nonadherence

- For patients with poor insight into illness or noncompliance with treatment, introduce peer support groups to help improve service user experience and quality of life (National Institute for Health and Care Excellence (NICE). Psychosis and schizophrenia in adults: prevention and management. CG178. London: NICE; 2014.)
- In-home intervention to prevent medical relapse, or readmission, and to enhance medication compliance
- For patients at risk of medication nonadherence, consider patient education and behavioral support initiatives such as text message reminders to take a medication and electronic medication dispensers.

Environmental exposure to toxic substances and other physical hazards

- For tobacco use and/or exposure information should be given about tobacco use cessation and the risk of second-hand smoke
- For lead poisoning, provide community resources for primary prevention and secondary prevention strategies (i.e., blood lead level testing and follow up) (Centers for Disease Control and Prevention, Lead Poisoning Prevention. 2019)

Unstable living environment or homelessness expected upon discharge

- Health insurance application
- Homeless shelter application and information should be given to patient
- Referral to walk-in state or federal sponsored outpatient programs
- Ensure access to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system (Pottie et al., CMAJ 2020, 192: E240-E54)
- For individuals with homelessness and the greatest service needs (i.e., serious mental illness, history of hospitalizations, functional impairment) consider Intensive Case Management Intervention, Assertive Community Treatment, or other Intensive Community-Based Treatments (Ponka et al., PLoS One 2020, 15: e0230896)

Unsafe home environment

- If home environment is unsafe, consider a skilled nursing facility (SNF)
- For patients at risk for falls, provide information on community-based fall prevention programs through the National Council on Aging

6:

Prior to initiating a review, obtain documentation of the following information in order to apply the criteria or provide additional information for a possible secondary review:

Clinical findings, including:

- Imaging results consistent with pneumonia
- Imaging results showing complicated pneumonia (i.e., empyema, parapneumonic effusion, lung abscess, necrotizing pneumonia, or multi-lobar involvement)
- O2 saturation
- Arterial PO_2
- Arterial or venous PCO_2
- Mental status
- Blood urea nitrogen level
- Respiratory rate
- Age
- Pregnancy
- Inability to absorb oral anti-infective
- Continued deterioration after outpatient anti-infective(s)
- Therapy

Any active comorbid conditions or risk factors, such as:

- Chronic obstructive pulmonary disease (COPD)
- Heart failure
- Mental illness
- Immunocompromised

Significant social or safety concerns, such as:

- The absence of a willing or able caregiver, if required
- An unsafe home environment
- Lack of access to medications and follow-up care
- Lack of access to transportation
- Non-compliance with treatments
- Frequent emergency department visits and/or hospital admissions

Interventions commonly not documented in the medical record, such as:

- Deep vein thrombosis (DVT) prophylaxis

7:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

8:

Radiology imaging should be performed on all patients hospitalized for the management of pneumonia to confirm diagnosis (given the inaccuracy of clinical signs and symptoms alone), to document the extent and severity of pulmonary infiltrates, and rule out complications that may warrant more intensive treatment (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67; Jokerst et al., J Am Coll Radiol 2018, 15: S240-s51). However, the diagnosis of pneumonia is not solely based on imaging findings due to the high inter-observer variability of chest x-ray readings

and the inability to distinguish viral from bacterial pneumonia in some cases. The diagnosis of pneumonia is a combination of clinical symptoms of lower respiratory tract infection (e.g., cough, fever, sputum production, dyspnea, and/or pleuritic chest pain) and the presence of a new infiltrate on imaging that cannot be explained by any other cause (Aliberti et al., Lancet 2021, 398: 906-19; Seo et al., Respiration 2019, 97: 508-17).
The radiologist reading the chest x-ray may not have access to the patient's clinical findings and may not specify a diagnosis of pneumonia when reporting imaging results. The chest x-ray findings are typically described in the report and may include a statement instructing the reader to correlate the findings to the clinical presentation.

9:

When a patient is admitted through the emergency department (ED), the initial dose of anti-infective is routinely administered while still in the ED. Guidelines support intravenous antibiotic therapy as the initial treatment for most patients requiring inpatient care, especially those with severe disease. There is a role for oral antibiotics in the inpatient setting to treat a subset of patients with less severe disease whose clinical findings still support admission, but the decision to administer oral antibiotics does not drive the need for inpatient treatment (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67; National Institute for Health and Clinical Excellence (NICE): Pneumonia in adults: diagnosis and management. Clinical Guideline 191. 2019; Lee et al., Jama 2016, 315: 593-602).

10:

Expected treatments for patients with bacterial pneumonia include anti-infective and oximetry or blood gas monitoring (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67; Cohoon et al., Am J Med 2018, 131: 307-16 e2; Kaysin and Viera, Am Fam Physician 2016, 94: 698-706; Jain et al., N Engl J Med 2015:[e-pub]).

11:

Imaging studies that can be utilized to aid in the diagnosis of pneumonia include chest x-ray, chest computed tomography (CT), and lung ultrasonography. Although a chest x-ray is the most commonly used imaging study, a chest CT may be required if the chest x-ray is negative but there is a high clinical suspicion for pneumonia (Aliberti et al., Lancet 2021, 398: 906-19). In addition, chest CTs have been found to be superior to chest x-rays in identifying lesions in the lung parenchyma as well as pleural abnormalities (Seo et al., Respiration 2019, 97: 508-17). The use of lung ultrasound is a newly emerging tool that has shown high accuracy in the diagnosis of pneumonia (Staub et al., J Emerg Med 2019, 56: 53-69).

12:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

13:

The Pneumonia Severity Index (PSI) is used to identify the risk of mortality in patients with community-acquired pneumonia (CAP) and considers the following factors:

- **Demographics**: Sex assigned at birth, Age, Nursing home resident
- **Comorbidities:** Cancer, liver or kidney, disease, heart failure, cerebrovascular disease
- **Physical Exam Findings:** Altered mental status, respiratory rate ≥ 30 breaths/min, heart rate ≥ 125 beats/min, systolic blood pressure < 90 mmHg, temperature < 95°F(35°C) or > 104°F(40°C)
- **Laboratory Results:** pH < 7.35, BUN > 10.7 mmol/L, sodium < 130 mEq/L, glucose > 13.9 mmol/L, hematocrit < 30%, PaO2 < 60 mm Hg, presence of pleural effusion

Point values are assigned to each factor and are summed to achieve a cumulative score, which in turn leads to an assigned risk class. The PSI Risk Classes are:

- Risk Class I - 0 points
- Risk Class II - ≤ 70 points
- Risk Class III - 71-90 points
- Risk Class IV - 91-130 points
- Risk Class V - > 130 points

Risk Classes I and II: low risk, outpatient treatment
Risk Class III: low risk, clinical factors and other factors (e.g., home environment) guide whether or not patient can be treated at home or in an observation setting

Risk Classes IV and V: moderate and high risk, respectively; inpatient treatment is recommended (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67; Fine et al., Arch Intern Med 1997; 157: 36-44).
PSI calculator

14:

In addition to clinical judgment, guideline standards recommend that clinicians use a validated clinical prediction rule to determine inpatient versus outpatient treatment location for adults with community-acquired pneumonia (CAP). Although both the Pneumonia Severity Index (PSI) and the CURB-65 are described in the guidelines, there is an expressed preference for the PSI (strong recommendation, moderate quality of evidence) over the CURB-65 (conditional recommendation, low quality of evidence). Despite a lower quality of evidence for the CURB-65, a conditional recommendation is offered by the guideline authors given its greater ease of use compared to the PSI (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67).

15:

The CURB-65 assessment is a clinical tool designed to predict the risk of 30-day mortality for patients with community-acquired pneumonia at the time of presentation. It consists of 5 variables (confusion, urea, respiratory rate, blood pressure, and age > 65), awarding 1 point for each variable present. For each additional clinical risk factor, the mortality rate increases. The CURB-65 criteria have also been validated to distinguish among those patients who can be safely managed at home, require short term observation care, or require inpatient admission. A score of 0 or 1 indicates low risk for mortality and suggests that a patient may be treated in the outpatient setting, a score of 2 indicates that a patient is at moderate risk for mortality and should be considered for short-stay inpatient treatment, and a score of 3 indicates a high risk for mortality and warrants hospital admission (Grief and Loza, Prim Care 2018, 45: 485-503; Lim et al., Thorax 2009; 64 Suppl 3: iii1-55).

16:

Confusion can be defined as a new onset disorientation to person, place, or time (Pieralli et al., Intern Emerg Med 2014, 9: 195-200). A score less or equal to 8 on the Abbreviated Mental Test (AMT) has also been used to define mental confusion in the CURB-65 score. The AMT score consists of 10 questions and each one is awarded a point if answered correctly. The questions are as follows (Lim et al., Thorax 2009, 64 Suppl 3: iii1-55):

- Age
- Date of birth
- Time (to the nearest hour)
- Year
- Hospital name
- Recognition of two people
- Recall address
- Date of a historical event
- Current president
- Count backwards from 20 to 1

17:

These criteria refer to the lack of response to treatment received through the medical practitioner's office, outpatient clinic (e.g., urgent care), teleconsultation, trial of home care services, emergency department (prior to decision to admit), or hospital Observation setting (if appropriate for this condition). These criteria can be applied when the patient fails to respond to more intensive treatment beyond the routine treatments or medications delivered for a chronic condition.

18:

Pneumonia treatment failure is defined as progressive pneumonia as evidenced by clinical deterioration and persistent pneumonia with no response to treatment after 72 hours of antibiotics. Pneumonia that does not respond to initial treatment has been associated with increased mortality (Grief and Loza, Prim Care 2018, 45: 485-503). Patients who are unable to tolerate or are noncompliant with the outpatient prescribed regimen are considered to have treatment failure since they are unable to be treated with a typical outpatient regimen.

19:

Patients experiencing continued deterioration of their infection demonstrate symptoms which may include fever, pain, mental status change, hypotension, hypoxia, nausea, and/or vomiting.

20:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

21:

Venous blood gases (VBG) may be used to assess a patient's ventilation or metabolic status. Venous pH or HCO_3 values closely approximate corresponding values derived from arterial blood gases (ABG) in conditions such as chronic obstructive pulmonary disease (COPD), respiratory distress syndrome, neonatal sepsis, renal failure, pneumonia, diabetic ketoacidosis, and status epilepticus. VBG should not be used to determine oxygenation. Patients with chronic CO_2 retention hospitalized for acute respiratory illness will exhibit increased PCO_2 levels. PCO_2 values should be compared with baseline values when available and other signs and symptoms of respiratory distress should be considered when determining the appropriate level of care (e.g., worsening dyspnea, high respiratory rate, decreased oxygen saturation, confusion, drowsiness) (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; McKeever et al., Thorax 2016, 71: 210-5).

22:

Complicated pneumonia refers to pneumonia with any of the following complications seen on imaging:

- Parapneumonic effusion
- Empyema
- Lung abscess
- Necrotizing pneumonia
- Multilobar, ≥ 2 lobes

23:

Patients who do not respond to antibiotics after 48-72 hours should undergo an evaluation (chest x-ray and lung ultrasound) for a parapneumonic effusion, empyema, necrotizing pneumonia, or lung abscess. Treatment is recommended at a tertiary care center with high-dose intravenous antibiotics if one of these complications are present on imaging (de Benedictis et al., Lancet 2020, 396: 786-98).

24:

Multilobar pneumonia places a patient at risk for treatment failure which has been associated with a longer length of stay and an increase in mortality (Peyrani et al., Chest 2020, 157: 34-41; Lim et al., Thorax 2009, 64 Suppl 3: iii1-55). Studies have also identified multilobar pneumonia as an independent risk factor for mortality, and as a result, is a common criterion in community-acquired pneumonia prognostic scoring tools (Smith et al., Ann Emerg Med 2021, 77: e1-e57; Mannu et al., Eur J Intern Med 2013, 24: 857-63).

25:

Frailty is a complex condition that occurs because of the aging process and makes individuals more vulnerable to stress, leading to adverse health outcomes like impaired mobility, falls, hospitalizations, and even mortality (Vahedi et al., Int J Geriatr Psychiatry 2022, 37: epub; Kwak and Thompson, Sports Med Health Sci 2021, 3: 1-10; Hoogendijk et al., Lancet 2019, 394: 1365-75). Studies have shown that older individuals identified as frail during hospital admission are at greater risk of in-hospital mortality, longer hospital stays, and functional decline at hospital discharge (Boucher et al., EClinicalMedicine 2023, 59: epub; Cunha et al., Ageing Res Rev 2019, 56: epub). Although there are many validated frailty screening tools available, there has yet to be one that is recognized as the gold standard for identifying frailty. Most of these tools are based on two models: the frailty phenotype and the deficit accumulation model. The frailty phenotype model focuses on the physical aspects of frailty (e.g., strength, walking speed, weight loss), while the deficit accumulation model identifies the number of deficiencies (e.g., functional, cognitive, comorbidities) to calculate a frailty index score (Hoogendijk et al., Lancet 2019, 394: 1365-75). However, both models have limitations in their implementation within the acute care setting, as the frailty phenotype assessment involves performing tests to evaluate the domains (e.g., grip strength, walking speed), and the deficit accumulation model is time-consuming. Therefore, frailty criteria have been developed based on literature and the two models mentioned above to help identify older adults who may have or be at risk for frailty.

26:

Hospitalized older individuals with frailty who have community-acquired pneumonia have been shown to have more severe disease, higher mortality rates, and an increased length of stay compared to non-frail individuals (Zhao et al., BMC Geriatr 2023, 23: 308; Luo et al., BMJ Open 2020, 10: e038370).

27:

Instruction: This criterion applies to patients who have previously been diagnosed with frailty and are currently experiencing the condition. This information can be retrieved from the medical record, such as the problem list or past medical history. However, describing a patient's appearance as "frail" in a progress note cannot be reviewed in this criterion.

28:

Instruction: The following criteria should be selected based on the patient's baseline or current health status. A patient's baseline frailty state can affect their risk for severe illness. Similarly, if a patient presents with signs of frailty in the setting of an acute illness or injury, it should also be seen as a marker of illness severity (Theou et al., Clin Geriatr Med 2018, 34: 369-86).

29:

Studies have suggested that incorporating cognitive impairment evaluation with physical characteristics enhances the predictability of frailty (Avila-Funes et al., J Am Geriatr Soc 2009, 57: 453-61). More recently, a systematic review and meta-analysis found that mean Mini-Mental State Examination (MMSE) scores were significantly lower in individuals who were identified as frail by the phenotype model, which focuses on physical function, compared to non-frail and pre-frail individuals. As a result of the strong relationship between physical frailty and cognitive impairment, the authors concluded that cognitive function should be incorporated into frailty assessments (Vahedi et al., Int J Geriatr Psychiatry 2022, 37: epub). Multiple frailty screening tools developed using the cumulative deficit approach assess the cognitive domain, specifically focusing on cognitive impairment or dementia (Mak et al., J Gerontol A Biol Sci Med Sci 2022, 77: 2311-9; Pajewski et al., J Gerontol A Biol Sci Med Sci 2019, 74: 1771-7; Clegg et al., Age Ageing 2016, 45: 353-60).

30:

Multimorbidity refers to the presence of 2 or more chronic conditions or diseases in an individual. Chronic conditions or diseases are those that last for a prolonged period (e.g., 1 year or more) or are permanent and result in disability, decreased quality of life, or require prolonged treatment or rehabilitation (Centers for Disease Control and Prevention, About Chronic Diseases. 2022 [cited Nov 8 2023]; Nicholson et al., J Clin Epidemiol 2019, 105: 142-6; Calderón-Larrañaga et al., J Gerontol A Biol Sci Med Sci 2017, 72: 1417-23). Examples of chronic conditions that have been identified in the literature include but are not limited to (Calderón-Larrañaga et al., J Gerontol A Biol Sci Med Sci 2017, 72: 1417-23):

- Chronic lung disease (e.g., COPD, asthma)
- Chronic kidney disease (e.g., stage 3-4)
- End-stage kidney disease
- Diabetes
- Hypertension
- Congestive heart failure
- Inflammatory bowel disease (e.g., Crohn's disease, ulcerative colitis)
- Mental health disorders (e.g., drug or alcohol dependence, depression, bipolar affective disorder)

31:

According to a recent systematic review and meta-analysis, 72%(0.72) of frail individuals have multimorbidity. The study also found a significant association between frailty and having two or more diseases (Vetrano et al., J Gerontol A Biol Sci Med Sci 2019, 74: 659-66).

32:

According to a study conducted at a Medicare Accountable Care Organization (ACO), the presence of one functional deficit on a frailty index tool in the electronic health record (EHR) was found to be independently associated with a higher risk of hospitalization and injurious falls (Pajewski et al., J Gerontol A Biol Sci Med Sci 2019, 74: 1771-7). When looking specifically at the association between hearing and vision deficits and frailty, a systematic review and meta-analysis found that hearing and vision impairment were associated with 2.5- and 3.0-fold higher odds of frailty, respectively (Tan et al., J Gerontol A Biol Sci Med Sci 2020, 75: 2461-70).

33:

Functional impairment refers to an individual's ability to independently perform activities of daily living (ADLs) (e. g., bathing, dressing, toileting, ambulating), instrumental activities of daily living (IADLs) (e.g., managing medications, finances, preparing meals, usage of the telephone, performing basic housework), or a deficit in hearing or vision that is not corrected by adaptive equipment (e.g., hearing aid, glasses).

34:

Malnutrition has been found to be closely associated with frailty (Lorenzo-López et al., BMC Geriatr 2017, 17: 108). This criterion can be met by one or more of the following:

- Documented weight loss, unintentional, of ≥ 10 pounds or ≥ 5%(0.05) of body weight in the past year (Fried et al., J Gerontol A Biol Sci Med Sci 2001, 56: M146-56)
- Score of ≤ 11 on the Mini Nutritional Assessment-Short Form (MNA-SF) (O'Caoimh et al., Int J Environ Res Public Health 2019, 16: epub; Lorenzo-López et al., BMC Geriatr 2017, 17: 108)
- Body mass index (BMI) < 18.5 kg/m^2 (Mak et al., J Gerontol A Biol Sci Med Sci 2022, 77: 2311-9; Pajewski et al., J Gerontol A Biol Sci Med Sci 2019, 74: 1771-7)
- Documented moderate or severe subcutaneous fat or muscle loss (Cederholm et al., Clin Nutr 2019, 38: 1-9; White et al., JPEN J Parenter Enteral Nutr 2012, 36: 275-83)

35:

Several studies have shown that individuals with polypharmacy are more likely to be frail (Gutiérrez-Valencia et al., Br J Clin Pharmacol 2018, 84: 1432-44). In a recent systematic review and meta-analysis of 66 studies, it was found that 59%(0.59) of older adults with frailty had polypharmacy, defined as the use of five or more medications at the same time. The study also revealed that the prevalence of polypharmacy was highest among older adults with frailty in the hospital setting, with a prevalence of 71%(0.71) (Toh et al., Ageing Res Rev 2023, 83: 101811).

36:

Chronic medications refer to prescription medications that are taken according to a schedule for at least 90 days, as opposed to being taken on an as-needed basis (Masnoon et al., BMC Geriatr 2017, 17: 230). This excludes topical medications, such as creams, or over-the-counter medications.

37:

Community-acquired pneumonia (CAP) can contribute to the worsening condition of patients with existing heart failure (HF) and chronic obstructive pulmonary disease (COPD). Previous studies have failed to show an association between pneumonia in COPD patients and an increase in mortality; however, a recent meta-analysis found that CAP in hospitalized COPD patients was associated with a longer length of stay, increased intensive care unit (ICU) admissions, and higher mortality (Yu et al., Respiration 2021, 100: 64-76). Patients with existing HF have been found to have up to a 4 times higher risk for mortality associated with pneumonia (Shen et al., J Am Coll Cardiol 2021, 77: 1961-73).

38:

The severity of chronic obstructive pulmonary disease (COPD) airflow limitations is classified into 4 grades based on the patient's baseline forced expiratory volume (FEV_1) measurement post-bronchodilator administration as follows (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024):

- GOLD 1 - Mild: FEV_1 ≥ 80%(0.80) predicted
- GOLD 2 - Moderate: FEV_1 50-79%(0.50-0.79) predicted
- GOLD 3 - Severe: FEV_1 30-49%(0.30-0.49) predicted

- GOLD 4 - Very severe: FEV_1 < 30%(0.30) predicted

39:

New York Heart Association (NYHA) classification for heart failure is defined as follows (Yap et al., Clinical cardiology 2015, 38: 621-8; New York Heart Association, Diseases of the Heart and Blood Vessels. Nomenclature and Criteria for diagnosis. 1964):

- Class I - No limitation of physical activity. Ordinary physical activity does not cause undue fatigue, palpitation, dyspnea, or anginal pain.
- Class II - Slight limitation of physical activity. Comfortable at rest. Ordinary physical activity results in fatigue, palpitation, dyspnea, or anginal pain.
- Class III - Marked limitation of physical activity. Comfortable at rest. Less than ordinary physical activity causes fatigue, palpitation, dyspnea, or anginal pain.
- Class IV - Inability to carry on any physical activity without discomfort. Symptoms of cardiac insufficiency or of anginal syndrome may be present even at rest. If any physical activity is undertaken, discomfort is increased.

40:

Heart failure (HF) patients with hypertension, diabetes mellitus, obesity, and metabolic syndrome require complex management. Treating comorbidities adequately can improve the outcomes of a HF exacerbation (Bozkurt et al., Circulation 2016, 134: e535-e78).

41:

The term "severe and persistent mental illness, cognitive impairment" also called "serious and persistent mental illness," refers to a mental illness or cognitive impairment that is characterized by severe symptoms of prolonged duration that require long-term treatment. The illness causes impaired functioning that interferes significantly with primary activities of daily living (ADLs) and results in an inability to maintain independent functioning without treatment, support, and rehabilitation for a prolonged or indefinite period of time. Examples include, but are not limited to, schizophrenia and other psychotic disorders, bipolar disorder, dementia, Alzheimer's disease, and severe cases of major depression.

42:

Patients hospitalized with heart failure, acute myocardial infarction, or pneumonia in addition to a psychiatric comorbidity have higher readmission rates within 30 days than those without a psychiatric diagnosis (Ahmedani et al., Psychiatr Serv 2015, 66: 134-40).

43:

Patient unreliable refers to the following:

- Inability or unwillingness to follow an outpatient monitoring or treatment plan
- Inability to understand the signs of deterioration and recognize the need for prompt medical attention

44:

Immunocompromised refers to the patient who has a decreased capacity to defend against invading organisms as a result of an impaired immune system due to several comorbidities. The principal manifestation of immunodeficiency is increased susceptibility to infection with unusual severity, frequency, and duration which differs from that of an individual who is immunocompetent who may be exposed to the same pathogen. Furthermore, individuals who are immunocompromised are increasingly prone to opportunistic infections. Immunodeficiency is classified as either primary (congenital), caused by inherited disorders of the immune system, or secondary (acquired), as a result of a disease process or as a result of treatment of other chronic diseases. Primary causes of immunodeficiency account for more than 200 genetic disorders that impact immune system function. Some examples of primary immunodeficiency include congenital immunodeficiency diseases (e.g., X-linked agammaglobulinemia), combined immunodeficiency disease (e.g., DiGeorge syndrome, Wiskott syndrome), and chronic granulomatous disease.
Examples of secondary immune deficiency include poorly controlled diabetes mellitus defined as HbA1c 9%(0.09) (drawn on admission or within the past three months), end-stage liver disease, hematopoietic malignancies, patients who have had allogeneic hematopoietic stem cell transplantation (HSCT), and patients receiving

immunosuppressive therapy (e.g., chemotherapy, radiation, anti-rejection medications, or prolonged high steroid use). Patients with severe, prolonged neutropenia (e.g., absolute neutrophil count less than 500/cu.mm(500x106/L), HSCT, human immunodeficiency virus (HIV) with CD4 count less than 200/cu.mm(200x106/L), a history of splenectomy, splenic dysfunction due to sickle cell disease, and those who have received intense chemotherapy (e. g., childhood acute myelogenous leukemia) are at greatest risk of infection (Centers for Disease Control and Prevention, National Diabetes Statistics Report. 2022; Bonilla et al., J Allergy Clin Immunol 2015, 136: 1186-205 e78; Lipska et al., Diabetes Care 2013, 36: 3535-42).

45:

The decision to hospitalize an immunocompromised patient with community-acquired pneumonia is a clinical decision, however, there should be a low threshold for admission since this population can appear stable on initial presentation but have the potential to deteriorate quickly (Ramirez et al., Chest 2020, 158: 1896-911).

46:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

47:

The Infectious Diseases Society of America/American Thoracic Society (IDSA/ATS) pneumonia severity criteria is a validated tool, with 3 or more minor criteria points defining severe pneumonia. Patients with pneumonia who are transferred to a higher level of care after admission have been found to have a higher mortality rate compared to patients who were directly admitted to the appropriate level of care form the emergency department; therefore, the use of the IDSA/ATS severity tool along with provider judgment is recommended when making a decision regarding medical appropriateness (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67).

48:

Instruction: The criterion WBC < 4,000/cu.mm(4 x10^9/L) is intended to be used for patients who have leukopenia due to infection alone (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67).

49:

Instruction: Hypotension is defined as a mean arterial pressure (MAP) < 65 mmHg or systolic blood pressure < 90 mmHg and < baseline. This line of criteria is intended for patients who obtain hemodynamic stability after fluid resuscitation. If a patient continues to have hypotension despite at least 2 liters of fluid resuscitation, the Critical level of care should be considered.

50:

High-flow nasal cannula (HFNC) delivers a fraction of inspired oxygen (FiO_2) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates up to 60 liters per minute, set independent from the FiO_2. The heated and humidified gas has been shown to improve comfort and aid in clearance of secretions. The high-flow rates reduce anatomical dead space by flushing out carbon dioxide from the upper airways, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. A Cochrane review was conducted in 2021 to determine the effectiveness of HFNC compared to standard oxygen therapy and noninvasive positive pressure ventilation (NIPPV) in adult patients requiring respiratory support in the intensive care unit (ICU). The systematic review found that HFNC had less treatment failure compared to standard oxygen therapy (i.e., face mask or nasal cannula with a flow rate of 15 L/min); however, it did not influence mortality rate or length of stay. When HFNC was compared to NIPPV, which included bilevel positive airway pressure (BiPAP) or continuous positive airway pressure (CPAP), there was no difference in treatment failure defined as intubation or re-intubation, mortality, or length of stay. In conclusion, HFNC has been shown to have less treatment failure compared to standard oxygen therapy and may be an alternative to NIPPV in patients treated for acute hypoxic respiratory failure (Lewis et al., Cochrane Database Syst Rev 2021, 3: Cd010172).

51:

The respiratory rate-oxygenation (ROX) index, defined as the ratio of oxygen saturation as measured by pulse oximetry (SpO_2)/fraction of inspired oxygen (FiO_2) to respiratory rate is a validated prediction tool that can be used to determine success of high flow nasal cannula (HFNC) in patients with acute hypoxic respiratory failure (AHRF) due to pneumonia. The ROX index can assist in the early identification of patients who are at high risk for intubation in order to prevent treatment delay (Roca et al., Am J Respir Crit Care Med 2019, 199: 1368-76).
A meta-analysis concluded that patients with a ROX index between 4.2 - 5.4 measured within 6 hours of HFNC initiation has the optimal confidence interval to predict HFNC outcome in patients with pneumonia related AHRF. Those with a ROX index less than 4.2 are likely at higher risk for HFNC failure, whereas those with a ROX index greater than 5.4 have a high chance for successful treatment with HFNC (Zhou et al., BMC Pulm Med 2022, 22: 121).

52:

Respiratory monitoring includes an evaluation of the patient's respiratory rate, heart rate, oxygen saturation, or work of breathing.

53:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).
The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.
Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

54:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO_2) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO_2 with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO_2 retention, where a precise inspired FiO_2 is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO_2. The following are estimates of O_2 delivered at the associated flow rates:
LOW-FLOW SYSTEMS
Nasal cannula
Room air 21%(*0.21*)
1L 24%(*0.24*) **4L** 36%(*0.36*)
2L 28%(*0.28*) **5L** 40%(*0.40*)
3L 32%(*0.32*) **6L** 44%(*0.44*)
Simple oxygen masks

- 35-50%(*0.35-0.50*) FiO_2 (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO_2 in the mask

Venturi masks

- 24-50%(*0.24-0.50*) FiO_2 (4-12 L/min)

Partial rebreathing masks

- 40-70%(*0.40-0.70*) FiO_2 (6-10 L/min)

Non-rebreathing masks
- 60-80%(*0.60-0.80*) FiO_2 (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS
Air-entrainment masks/nebulizers
- 24-40%(0.24-0.40) FiO_2

Heated, humidified high-flow concentrating nasal cannula (HFNC)
- 24-100%(0.24-1.0) FiO_2

55:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

56:

Extracorporeal membrane oxygenation (ECMO) or extracorporeal life support (ECLS) provides oxygen and removes carbon dioxide from blood. ECMO/ECLS facilitates the recovery of cardiopulmonary function when other measures fail to improve gas exchange. Indications for the use of ECMO/ECLS in children include refractory septic or cardiogenic shock, cardiac arrest as a bridge to transplant, or refractory hypoxemic or hypercarbic respiratory failure. There is clear evidence to support the use of ECMO/ECLS in the pediatric population, but definitive studies in adults are lacking. However, in adults, observational studies and other data support the use of ECMO/ECLS for severe refractory, hypoxemic, or hypercarbic respiratory failure (Mehta et al., American College of Cardiology Perspectives and Analysis 2018; Napp et al., Herz 2017, 42: 27-44; Guirand et al., J Trauma Acute Care Surg 2014, 76: 1275-81). ECMO/ECLS incorporates the use of an external circuit, a cannula for vascular access, a pump to propel the blood through the circuit, and an artificial membrane that oxygenates the blood. Common complications include hemorrhage, hemolysis, and infection. Stroke, embolism, and seizures in children have also been reported. Infants with intraventricular hemorrhage (IVH) or those at high risk for IVH (i.e., weight less than 2000 grams, age less than 35 weeks) are not candidates for this therapy (Abrams et al., J Am Coll Cardiol 2014, 63: 2769-78; Domico et al., Pediatr Crit Care Med 2012, 13: 16-21; MacLaren et al., Intensive Care Med 2012, 38: 210-20; Peek et al., Health Technol Assess 2010, 14: 1-46; Mugford et al., Cochrane Database Syst Rev 2008: CD001340).

57:

Fluid resuscitation for patients with sepsis-induced hypoperfusion or hypotension usually begins with 30 mL per kg in adults; the recommendation for this starting amount was downgraded from a strong to a weak recommendation in the 2021 Surviving Sepsis Campaign Guidelines as a change from the 2016 Guidelines (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Semler and Rice, Clin Chest Med 2016, 37: 241-50). Traditionally, parameters used to assess response to fluid administration included improvements in mean arterial pressure (MAP), urine output, or central or mixed venous oxygen saturation. Surviving Sepsis Campaign guidelines recommend that decisions regarding additional fluid administration should be based on frequent reassessment of hemodynamic and clinical status, as well as dynamic variables of fluid responsiveness including inferior vena cava diameter, pulse pressure variation, stroke volume variation, and lactate clearance (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552).
Patients with ongoing hypotension and inability to tolerate the recommended amount of fluid due to volume overload may need early initiation of vasopressor therapy.

58:

Patients with sepsis-induced tissue hypoperfusion can present with elevated blood lactate levels greater than 2 mmol/L(18.0 mg/dL) (Shankar-Hari et al., JAMA 2016, 315: 775-87).

59:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

60:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

61:

Instruction: The criteria point "Comorbid pneumonia in a hospitalized patient" can only be applied to patients who have developed pneumonia as a complication of their current inpatient stay in an acute care facility. The criteria points cannot be applied for patients who have been either discharged to home or transferred to a post-acute facility.

62:

The CMS Two-Midnight Rule generally requires that the medical record support a patient’s need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.
For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.
NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.
(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

63:

The NICE guidelines recommend that patients with community-acquired pneumonia remain hospitalized if any 2 of the following are present (National Institute for Health and Clinical Excellence (NICE), Pneumonia (community-acquired): antimicrobial prescribing. 2019):

- Temperature > 37.5°C (99.5°F)
- Respiratory rate ≥ 24 breaths per minute
- Heart rate > 100 beats per minute
- Systolic blood pressure ≤ 90 mmHg
- O_2 sat < 90%(0.90) on room air
- Abnormal mental status
- Inability to eat without assistance

64:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

65:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

66:

Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

67:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirement. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit.

68:

Routes for nutrition may be by mouth (PO), intravenously (IV), jejunostomy tube (J-tube), or gastrostomy tube (G-tube).

69:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

70:

Intravenous antibiotics should be switched to an oral regimen as soon as there is clinical stability, defined as normal vital signs (heart rate, respiratory rate, blood pressure, oxygen saturation, and temperature), ability to eat, and normal mentation. Most patients achieve clinical stability in 48-72 hours after the initiation of intravenous antibiotics (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67).

71:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

72:

Chest tubes are used to evacuate air, fluid, or blood from the pleural space. For malignancies involving the pleura, chest tubes may also be used for the administration of sclerosing agents. Serial chest x-rays may be performed to ensure that there is no re-accumulation of air or fluid. The chest tube may be removed when there is no evidence of air leak, output has slowed to 250-400 mL/day, and the lung is fully expanded (Vuong et al., Respir Med 2018, 137: 152-66).

73:

Swallowing assessment includes direct nursing or physician observation while swallowing, or a formal swallowing evaluation by a speech-language pathologist.

74:

Instruction: These criteria apply to the administration of a bronchodilator via a nebulizer, an inhaler, or an intravenous route. According to the available evidence, inhalers produce outcomes that are at least equivalent to nebulizers. Further research is needed to determine the optimal inhalation device according to individual patient characteristics and disease state. Education on correct usage is vitally important to ensure effective medication delivery to the lungs (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; van Geffen et al., Cochrane Database Syst Rev 2016: CD011826).

75:

Functional impairment refers to any condition that results in activity limitations impacting the patient's ability to ambulate, transfer, and/or perform activities of daily living (ADLs) or instrumental activities of daily living (IADLs). In pediatric patients, functional impairment includes limitations that impact the patient's developmental progress, such as deficits in mental status, sensory functioning, communication, motor functioning, feeding, and respiratory status. The patient will require assistance with one or more limitations.

76:

Instruction: These criteria refer to a functional impairment that develops during the course of this hospitalization.

77:

Rehabilitation may include, but is not limited to, physical therapy, occupational therapy, or speech-language pathology.

78:

Management of patients admitted with a heart failure (HF) exacerbation may include supplemental oxygen, diuretic, guideline-directed medical therapy (GDMT), which includes angiotensin-converting enzyme inhibitor (ACEi) or angiotensin receptor blocker (ARB) or angiotensin receptor neprilysin inhibitor (ARNi), beta-blocker, mineralocorticoid receptor antagonists (MRA), and sodium-glucose cotransporter 2 inhibitors (SGLT2i), cardiac monitoring, and deep vein thrombosis (DVT) prophylaxis (Maddox et al., J Am Coll Cardiol 2021, 77: 772-810).

79:

Bilateral pulmonary infiltrates seen on chest x-ray are consistent with the diagnosis of left-sided heart failure.

80:

In patients hospitalized with acute heart failure (HF) due to an increased risk of atrial and ventricular arrhythmias, continuous cardiac monitoring is recommended. Monitoring may be discontinued when there is no evidence of a hemodynamically significant arrhythmia and the symptoms of acute HF have improved (Sandau et al., Circulation 2017, 136: e273-e344).

81:

Intravenous (IV) loop diuretics (e.g., furosemide, torsemide, bumetanide, ethacrynic acid) are administered to relieve the symptoms of dyspnea and congestion without excessively reducing intravascular volume. Due to their relatively short half-life, diuretic effectiveness can be enhanced by continuous administration or multiple boluses daily. Careful monitoring of daily weights, orthostatic vital signs, intake and output, electrolytes, and renal function are key components of diuretic therapy. If a patient does not initially respond to IV diuretics, other options may be considered, including increasing the dose to ensure adequate drug levels reach the kidneys and adding a second diuretic, typically a thiazide, to the loop diuretic in order to improve diuretic responsiveness (Heidenreich et al., Circulation 2022, 145: e895-e1032; Maddox et al., J Am Coll Cardiol 2021, 77: 772-810; Rosendorff et al., Circulation 2015).

82:

Instruction: For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

83:

Volume expanders are fluids administered intravenously to increase circulatory volume. Studies have demonstrated that a balanced crystalloid solution (e.g., Lactated Ringer's) is preferable to colloids in restoration of intravascular volume in sepsis and septic shock (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Winters et al., J Emerg Med 2017, 53: 928-39). However, crystalloids have been found to be less effective than colloids at stabilizing hemodynamic endpoints in critically ill patients (Martin and Bassett, J Crit Care 2019, 50: 144-54). The type of fluid selected for administration should be based on the indication for its use and other patient-specific factors.
Instruction: In order to apply criteria for volume expanders, there should be documentation of a volume deficit supported by clinical findings. The volume of infusion is patient-specific and varies based on the cause of volume depletion, comorbid condition, and patient response. Criteria for volume expander should not be applied for maintenance intravenous fluids or electrolyte replacement.

84:

This criteria point can be satisfied with the weight-based amount specified, or, for an average-size adult patient (~70kg), by the administration of at least 2 liters of crystalloid, a colloid equivalent (e.g., two 250 ml bottles of 5% albumin), or a combination of crystalloid and colloid (at least 1 liter of crystalloid and at least one 250 ml bottle of 5% albumin) (Ernest et al., Crit Care Med 1999, 27: 46-50).
Patients who require less than 30 ml/kg of intravenous fluids in 24 hours and do not meet other criteria for the Intermediate level of care may be reviewed at the Acute level of care; other patients who are not able to tolerate the specified amount of fluid and may require early initiation of vasopressors should be reviewed at the Critical level of care.

85:

The CMS Two-Midnight Rule generally requires that the medical record support a patient’s need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.
For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.
Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status;** those who do not meet criteria would be appropriate for secondary review.
(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

86:

Expected progress after 72h

- O_2 saturation at or close to baseline
- Afebrile
- Laboratory values improving
- Activities at or close to baseline
- Preparing for discharge within the next 24 hours.

If no, consider:

- Review of sputum and culture results, including sensitivities
- Repeat blood cultures and laboratory tests
- Repeat imaging such as a chest tomography to assess for complicated pneumonia or pulmonary embolism
- Respiratory therapy for ineffective airway clearance
- Speech therapy consult to assess for aspiration
- Antibiotic regimen change for persistent fever, leukocytosis, dyspnea, decreased O_2 saturation, or worsening clinical status

87:

Functional impairments may be safely managed in the home environment when the patient is willing and able to perform self-care independently with or without assistive devices, has sufficient caregiver support available to adequately meet the patient's needs or home care services are arranged.

88:

Discharge to home:
The patient or caregiver should be able to safely manage impairment at home.
Discharge home with services:
Home care services may need to be arranged if the functional impairment cannot be safely managed by the patient or caregiver in the home setting.
Discharge to inpatient facility:
If the patient's functional impairment cannot be safely managed at home or requires services beyond home care, an inpatient rehabilitation stay should be considered.

89:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

90:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

91:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

92:

Smoking cessation should be a goal of smokers who are hospitalized with pneumonia, as smoking is an independent risk factor for the development of pneumonia (Aliberti et al., Lancet 2021, 398: 906-19; Mandell et al., Clin Infect Dis 2007; 44 Suppl 2: S27-72).

93:

Patients with pneumonia who are being discharged should review influenza and pneumococcal vaccinations. If the patient has not received either vaccination prior to discharge, arrangements should be made for outpatient administration (Mandell et al., Clin Infect Dis 2007; 44 Suppl 2: S27-72; Fiore et al., MMWR Recomm Rep 2010, 59: 1-62; Centers for Disease Control and Prevention. MMWR Recomm Rep 2010, 59: 1102-6; Tomczyk et al., MMWR Morb Mortal Wkly Rep 2014, 63: 822-5).

94:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

95:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

96:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

97:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

98:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

99:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

100:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

101:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

102:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

103:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

104:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting

- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

105:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

106:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.
The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

107:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

108:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

109:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

110:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

111:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.